

Lucas Root

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Computer Science student (Cybersecurity & AI threads) at the Georgia Institute of Technology, transferring from the University of Kansas, seeking a software engineering internship.

EDUCATION

Georgia Institute of Technology <i>B.S. in Computer Science: Cybersecurity & AI</i>	Atlanta, GA Aug. 2026 – Expected May 2028
University of Kansas <i>B.S. in Computer Science & Cybersecurity Engineering, Honors College</i> – 3.9/4.0 GPA — Coursework: Programming 2, Software Engineering, Discrete Structures, Embedded Systems	Lawrence, KS Aug. 2024 – May 2026
Menlo-Atherton High School <i>High School Diploma, 4.08 GPA</i>	Menlo Park, CA May 2024

EXPERIENCE

VICEROY MAVEN Research Intern <i>Air Force Research Laboratory (AFRL/RI), U.S. Department of Defense — Griffiss Institute</i> – Contributed to four AFRL research initiatives: privacy-preserving retrieval (SWEEP, SHADOW), video annotation & inter-annotator agreement analysis (SCOPE), and real-time streaming video-language modeling (RANGER) — using Python, OHTTP/ODoH, mitmproxy, local LLM integration, and Krippendorff's Alpha.	Jun. 2026 – Aug. 2026 Rome, NY
Founder & Lead Developer, Joumé <i>Clothing brand — design, dev, sourcing, finances</i> – Founded and run a clothing brand end-to-end: storefront (Rust, SQLite, Stripe), design, sourcing, and finances.	Dec. 2024 – Present Menlo Park, CA
Capture The Flag Challenges <i>OverTheWire, VICEROY ShieldBearer & Sentinel Watch</i> – Active in cybersecurity challenges (web exploitation, network recon, Linux privilege escalation, OverTheWire): placed 9th of 44 in VICEROY ShieldBearer, 8th in Sentinel Watch.	Dec. 2024 – Present Online
Navy Midshipman <i>US Navy ROTC, University of Kansas</i> – Completed New Student Indoctrination and Orientation; recognized for leadership potential.	Aug. 2024 – Dec. 2024 Lawrence, KS
Waiter / Busboy <i>Dessert Republic</i> – Provided customer service in a bilingual (English/Cantonese) environment at a family-owned restaurant.	Jun. 2022 – Nov. 2022 San Mateo, CA

PROJECTS

TURT: Custom Unix Shell 🔗 <i>C, Linux System Calls, Makefile</i> – Engineered a POSIX-compliant shell in C, managing process execution via <code>fork</code> , <code>execvp</code> , and <code>wait</code> . – Built a custom tokenizer and signal handling for built-in commands, replicating core Bash functionality (verified via Valgrind).	Oct. 2025 - Present
Praxis: Student Social Network 🔗 <i>Rust, Next.js, Postgres</i> – Architecting a full-stack platform connecting students for project collaboration, with a Rust backend for type safety and low-latency queries. – Designing a relational database schema in PostgreSQL to manage user relationships and project feeds.	Jan. 2026 - Present
Syllabus2Cal 🔗 <i>TypeScript, Next.js, OpenAI API</i> – Built a tool converting unstructured syllabus text into structured .ics data using LLM function calling. – Optimized API usage costs and response times through effective prompt engineering and caching.	Jan. 2025 - Present

LEADERSHIP & AWARDS

President, Kansas Crew <i>University of Kansas</i> – Lead logistics, recruitment, and community partnerships for the official university rowing club.	Aug. 2024 – Present Lawrence, KS
Varsity Rower, NorCal Crew <i>Redwood City, CA</i> – Competed nationally (13th at 2023 Head of the Charles); coached novice rowers ages 11–15.	Aug. 2019 – May 2024
KU Excellence Scholarship <i>University of Kansas</i>	2024

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, Java, C#, JavaScript/TypeScript, SQL
Developer Tools: Git/GitHub, Docker, Linux/Unix, GDB, Valgrind, Make, VSCode
Concepts: Memory Management, OS Architecture, Embedded Systems, Data Structures